

PATHUMTHANI, Thailand

WMO: 484190

Lat: 14.1162N

Lon: 100.6203E

Elev: 23

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	66.5	69.4	54.5	63.2	71.3	58.0	72.0	74.0	11.8	81.6	10.0	82.1	3.3	10	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	16.0	99.2	81.8	97.7	81.3	96.3	80.9	84.8	95.1	83.7	92.9	82.8	91.6	6.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.4	169.0	90.7	81.5	163.7	89.0	80.6	159.3	87.5	49.2	95.3	47.8	93.0	46.8	91.7	96.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature															
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years												
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max										
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)								
12.6	11.1	9.6	61.7	100.7	3.6	2.1	59.2	102.2	57.1	103.4	55.1	104.6	52.5	106.1								
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB								
			61.7	56.9	100.7	86.2	3.6	3.9	2.1	3.8	59.2	54.1	102.2	88.9	57.1	51.8	103.4	91.2	49.6	93.3	46.8	96.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.8	80.5	83.6	86.0	87.9	87.7	86.4	85.4	85.3	84.7	84.7	83.8	81.1
	DBStd	3.60	3.83	2.76	3.38	3.05	2.94	2.02	2.25	1.91	2.05	2.32	2.83	3.92
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12686	946	941	1115	1138	1169	1091	1099	1095	1041	1075	1013	963
	CDD65	7211	481	521	650	688	704	641	634	630	591	610	563	498
	CDH74	89169	5129	6328	8538	9491	9474	8321	7899	7713	7001	7278	6623	5374
	CDH80	42153	2072	2958	4420	5350	5114	4098	3624	3456	2954	3104	2832	2171
Wind	WSAvg	3.6	2.7	3.2	4.0	4.1	4.1	4.5	4.4	4.4	3.7	2.6	3.0	3.1
Precipitation	PrecAvg	47.5	0.4	0.4	1.5	3.0	6.1	5.2	5.3	6.2	10.7	7.2	1.5	0.2
	PrecMax	57.5	2.0	1.3	4.9	7.8	12.1	8.9	10.0	9.9	15.3	17.2	5.7	1.7
	PrecMin	34.8	0.0	0.0	0.1	0.3	1.5	1.1	2.4	3.3	6.6	1.6	0.0	0.0
	PrecStd	6.4	0.5	0.4	1.5	2.0	2.4	1.7	1.9	1.7	2.6	2.8	1.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.4	96.4	99.6	101.6	101.1	98.1	96.8	96.1	95.8	96.3	95.6	95.3
		MCWB	78.3	80.2	80.4	82.0	83.3	81.8	81.8	80.4	81.2	81.3	80.1	79.8
	2%	DB	92.4	94.7	97.4	99.7	98.7	95.8	94.6	94.0	93.8	94.0	93.7	93.3
		MCWB	77.1	78.6	80.1	82.0	82.1	81.6	80.5	80.0	80.6	80.6	79.6	78.6
	5%	DB	90.7	93.3	95.8	98.0	96.8	94.0	92.8	92.4	92.2	92.4	92.3	91.4
		MCWB	76.4	78.2	79.9	81.5	81.7	81.0	80.1	79.7	80.2	80.3	79.2	77.6
	10%	DB	89.0	91.6	94.0	96.2	94.9	92.3	91.2	91.1	90.7	90.9	90.7	89.3
		MCWB	75.9	77.5	79.6	81.1	81.2	80.3	79.6	79.4	79.8	80.0	78.7	76.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	82.8	84.1	86.8	85.8	84.9	84.0	82.5	82.8	83.0	82.4	81.5
		MCDB	89.8	94.0	95.6	98.3	97.7	94.5	93.7	90.2	92.0	91.9	91.5	92.3
	2%	WB	79.8	81.0	82.5	85.3	84.4	83.6	82.5	81.5	81.9	81.9	81.4	80.1
		MCDB	88.5	91.0	93.0	95.8	94.6	91.8	90.4	89.2	90.5	90.3	90.1	89.9
	5%	WB	78.9	80.0	81.5	83.6	83.4	82.3	81.2	80.8	81.1	81.3	80.7	79.2
		MCDB	87.0	89.4	91.5	92.6	92.6	89.5	88.7	88.6	89.0	89.2	89.0	88.1
	10%	WB	77.9	79.1	80.7	82.3	82.4	81.4	80.4	80.1	80.5	80.7	80.0	78.2
		MCDB	85.4	87.9	90.2	91.4	90.8	88.6	87.8	87.8	87.8	88.1	87.5	86.2
Mean Daily Temperature Range	5% DB	MDBR	16.7	17.0	16.4	16.0	14.0	12.6	11.8	12.2	12.0	12.2	13.4	15.4
		MCDBR	18.1	18.4	18.5	17.9	16.1	14.8	13.8	13.9	14.2	14.3	15.2	16.7
		MCWBR	5.9	5.3	5.0	5.3	4.8	4.3	4.1	4.0	4.5	4.3	4.7	5.5
	5% WB	MCDBR	15.7	16.7	17.2	17.1	15.3	13.7	12.8	13.0	13.1	13.2	13.8	15.7
		MCWBR	5.8	5.4	5.6	6.3	5.6	5.0	4.7	4.5	4.8	4.5	4.6	5.6
Clear-Sky Solar Irradiance	taub		0.540	0.616	0.651	0.589	0.507	0.467	0.458	0.459	0.487	0.576	0.518	0.506
	taud		1.954	1.801	1.753	1.937	2.220	2.330	2.358	2.341	2.254	1.964	2.100	2.075
	Ebn at Noon		240	228	224	238	255	262	265	268	262	235	245	246
	Edh at Noon		57	69	75	62	46	41	40	41	44	58	49	50
All-Sky Solar Radiation	RadAvg		1584	1714	1776	1909	1767	1682	1596	1597	1550	1528	1563	1540
	RadStd		92	69	126	108	123	89	78	103	84	86	97	69

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	+0.49	+0.50	N/A	N/A	N/A	N/A

Nomenclature: See separate page